

Deep Generative Models: Linear Dynamical Systems

Fall Semester 2025

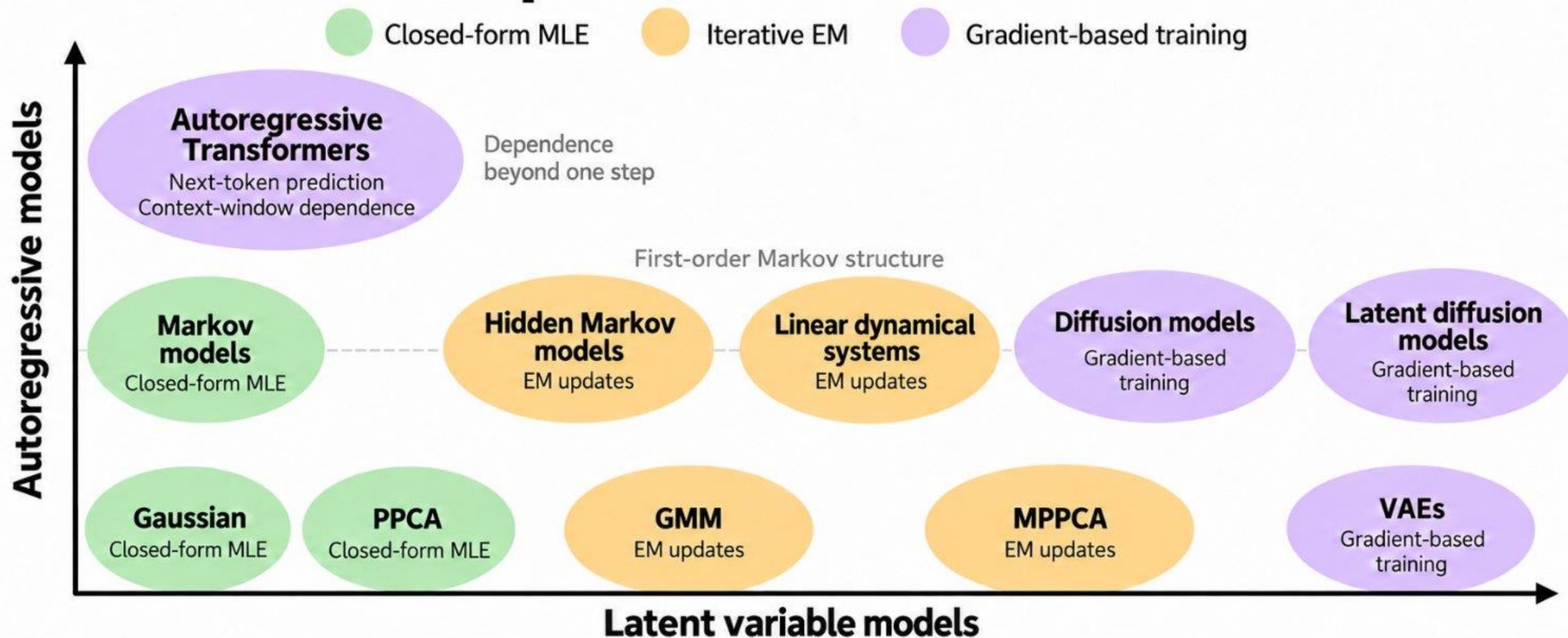
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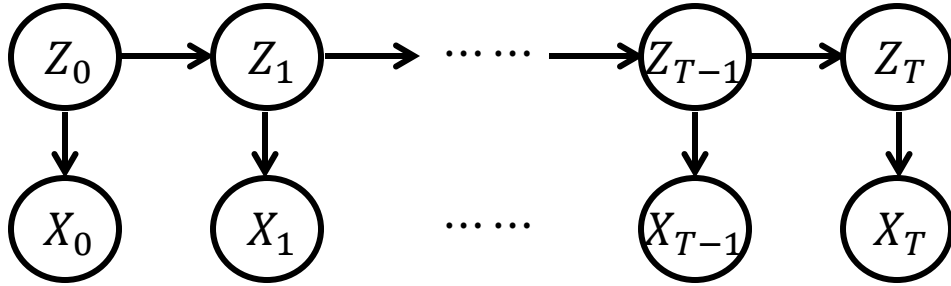
Taxonomy of Generative Models

A Map of Generative Models



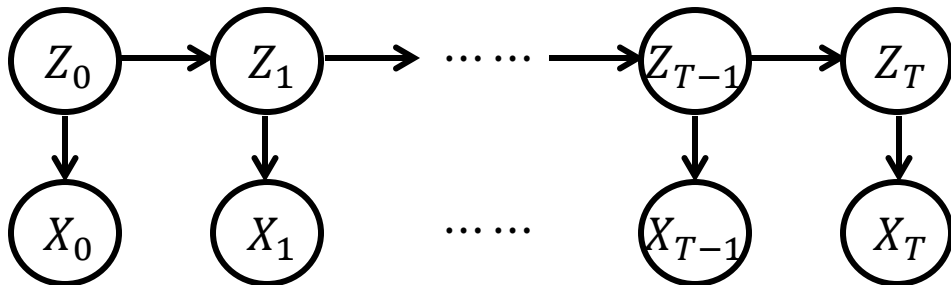
HMMs and Linear Dynamical Systems

Hidden Markov Models (HMMs)



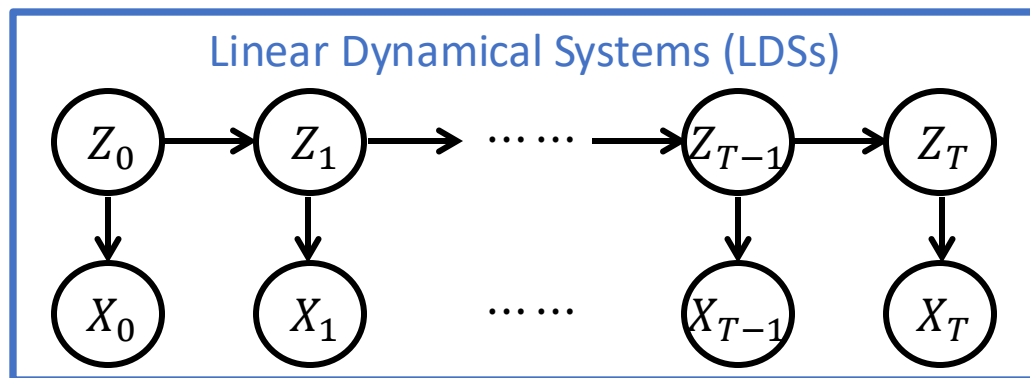
- Hidden state Z_t and observation X_t are **discrete random variables**
- State transition and emission probabilities are **categorical**

Linear Dynamical Systems (LDSs)



- Hidden state Z_t and observation X_t are **continuous random vectors**
- State transition and emission probabilities are **Gaussian**

Linear Dynamical Systems



- Hidden state Z_t and observation Y_t are **continuous random vectors**
- State transition and emission are **Gaussian**

Model Parameters:

- $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

d : state dimension

D : output dimension

$$\mathbf{S}_0 \in \mathbb{R}^{d \times d}$$

$$\mathbf{S}_0 \succ 0$$

$$\mathbf{A} \in \mathbb{R}^{d \times d}$$

$$\mathbf{C} \in \mathbb{R}^{D \times d}$$

$$\mathbf{Q} \in \mathbb{R}^{d \times d}$$

$$\mathbf{Q} \succ 0$$

$$\mathbf{R} \in \mathbb{R}^{D \times D}$$

$$\mathbf{R} \succ 0$$

- Initial Distribution:

$$p(Z_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

- State Transition:

$$p(Z_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

- State Emission:

$$p(X_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

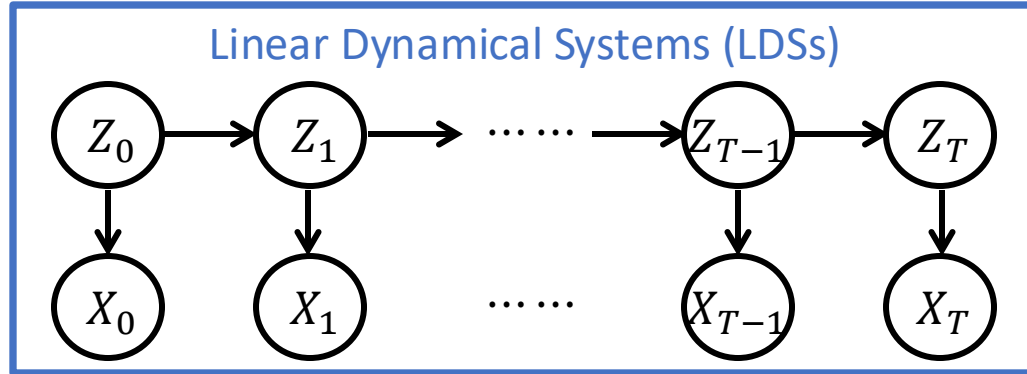
We assume $\mathbf{w}_t, \mathbf{v}_t$ are independent from each other and independent from $\mathbf{z}_0$

	Probabilistic Description	Algebraic Description
State Transition	$p(Z_t \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$	$Z_t = \mathbf{A}Z_{t-1} + \mathbf{w}_t$ with $\mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q})$
State Emission	$p(X_t \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$	$X_t = \mathbf{C}Z_t + \mathbf{v}_t$ with $\mathbf{v}_t \sim \mathcal{N}(0, \mathbf{R})$

Filtering and Smoothing

- **P1: Filtering.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_t)$, infer the current state $\mathbf{z}_t$, that is to compute
$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$
 - e.g., what is the current state of the missile given its position over some past time?
- **P2: Smoothing.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, infer the past state $\mathbf{z}_t$, that is to compute
$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_T)$$
 - e.g., where did the missile originate given we observed it over some time?
- **Remark.** You may find **P1** and **P2** familiar
 - In HMMs, we solved them via recursively updating $\alpha_j(t), \gamma_j(t)$

Basic Properties



- Hidden state Z_t and observation Y_t are **continuous random vectors**
- State transition and emission are **Gaussian**

Model Parameters:

- $\theta := (\pi_0, \Sigma_0, A, Q, C, R)$

d : state dimension

D : output dimension

$$\Sigma_0 \in \mathbb{R}^{d \times d}$$

$$\Sigma_0 \succ 0$$

$$A \in \mathbb{R}^{d \times d}$$

$$C \in \mathbb{R}^{D \times d}$$

$$Q \in \mathbb{R}^{d \times d}$$

$$Q \succ 0$$

$$R \in \mathbb{R}^{D \times D}$$

$$R \succ 0$$

- Joint distribution is Gaussian:

$$p_{\theta}(\mathbf{x}_0, \dots, \mathbf{x}_T, \mathbf{z}_0, \dots, \mathbf{z}_T) = \underbrace{p_{\theta}(\mathbf{z}_0)}_{\text{Gaussian}} \prod_{t=0}^T \underbrace{p_{\theta}(\mathbf{x}_t | \mathbf{z}_t)}_{\text{Gaussian}} \prod_{t=1}^T \underbrace{p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1})}_{\text{Gaussian}}$$

- Therefore, “any conditional distribution” is Gaussian
 - Vague, but look at your question 1 of homework 1 (next page)

Gaussian Conditioning (HW 1)

- If $\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{bmatrix} \right)$, then the conditional distribution $p(\mathbf{a} \mid \mathbf{b})$ is Gaussian with mean $\boldsymbol{\mu}_{a|b}$ and covariance $\boldsymbol{\Sigma}_{a|b}$ given by

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} (\mathbf{b} - \boldsymbol{\mu}_b)$$

original mean & variance of $\mathbf{a}$

correction upon observing $\mathbf{b}$

$$\boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} \boldsymbol{\Sigma}_{ba}$$

Gaussian Combining (Extension of Problem 1a, HW 1)

Gaussian Combining: If $\mathbf{x} = \mathbf{C}\mathbf{z} + \mathbf{v}$ with $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$ and $\mathbf{v} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_v)$ then

$$\begin{bmatrix} \mathbf{z} \\ \mathbf{x} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_z \\ \mathbf{C}\boldsymbol{\mu}_z \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_z & \boldsymbol{\Sigma}_z \mathbf{C}^\top \\ \mathbf{C}\boldsymbol{\Sigma}_z & \mathbf{C}\boldsymbol{\Sigma}_z \mathbf{C}^\top + \boldsymbol{\Sigma}_v \end{bmatrix} \right)$$

- **Proof:** The proof is finished by computing the following quantities:

- $\mathbb{E}[\mathbf{x}] = \mathbb{E}[\mathbf{C}\mathbf{z} + \mathbf{v}] = \mathbf{C}\mathbb{E}[\mathbf{z}] + 0 = \mathbf{C}\boldsymbol{\mu}_z$
- $\text{Cov}(\mathbf{z}, \mathbf{x}) = \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu}_z)(\mathbf{x} - \mathbf{C}\boldsymbol{\mu}_z)^\top] = \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu}_z)(\mathbf{C}\mathbf{z} + \mathbf{v} - \mathbf{C}\boldsymbol{\mu}_z)^\top] = \boldsymbol{\Sigma}_z \mathbf{C}^\top$
- $\begin{aligned} \text{Cov}(\mathbf{x}) &= \mathbb{E}[(\mathbf{x} - \mathbf{C}\boldsymbol{\mu}_z)(\mathbf{x} - \mathbf{C}\boldsymbol{\mu}_z)^\top] = \mathbb{E}[(\mathbf{C}\mathbf{z} + \mathbf{v} - \mathbf{C}\boldsymbol{\mu}_z)(\mathbf{C}\mathbf{z} + \mathbf{v} - \mathbf{C}\boldsymbol{\mu}_z)^\top] \\ &= \mathbb{E}[\mathbf{C}(\mathbf{z} - \boldsymbol{\mu}_z)(\mathbf{z} - \boldsymbol{\mu}_z)^\top \mathbf{C}^\top + \mathbf{C}(\mathbf{z} - \boldsymbol{\mu}_z)\mathbf{v}^\top + \mathbf{v}(\mathbf{z} - \boldsymbol{\mu}_z)^\top \mathbf{C}^\top + \mathbf{v}\mathbf{v}^\top] \\ &= \mathbf{C} \cdot \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu}_z)(\mathbf{z} - \boldsymbol{\mu}_z)^\top] \cdot \mathbf{C}^\top + 0 + 0 + \mathbf{R} = \mathbf{C}\boldsymbol{\Sigma}_z \mathbf{C}^\top + \mathbf{R} \end{aligned}$

Filtering and Smoothing

- **P1: Filtering.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_t)$, compute

$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **P2: Smoothing.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, compute

$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_T)$$

- Since $p_{\theta}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$ is Gaussian ($\forall s, t$), it suffices to compute

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$

$$\hat{\Sigma}_{s|t} := \mathbb{E} \left[(\mathbf{z}_s - \hat{\mathbf{z}}_{s|t})(\mathbf{z}_s - \hat{\mathbf{z}}_{s|t})^{\top} \mid \mathbf{x}_0, \dots, \mathbf{x}_t \right] = \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- We will do so recursively (first for filtering and then for smoothing)

Filtering: Compute $\hat{\mathbf{z}}_{0|0}, \hat{\Sigma}_{0|0}$

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

- **P1: Filtering.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_t)$, compute $p_\theta(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$
- Let's begin with the simplest case:
 - What are the mean $\hat{\mathbf{z}}_{0|0} = \mathbb{E}[\mathbf{z}_0 \mid \mathbf{x}_0]$ and covariance $\hat{\Sigma}_{0|0} = \text{Cov}(\mathbf{z}_0 \mid \mathbf{x}_0)$ of $\mathbf{z}_0$ given $\mathbf{x}_0$?
- **High-level Idea.**
 1. find the mean and covariance of $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$ via **Gaussian combining**
 2. find the mean $\hat{\mathbf{z}}_{0|0}$ and covariance $\hat{\Sigma}_{0|0}$ of $\mathbf{z}_0 \mid \mathbf{x}_0$ via **Gaussian conditioning**

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} (\mathbf{b} - \boldsymbol{\mu}_b), \quad \boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab} \boldsymbol{\Sigma}_{bb}^{-1} \boldsymbol{\Sigma}_{ba}$$

Filtering: Compute $\hat{\mathbf{z}}_{0|0}, \hat{\Sigma}_{0|0}$

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **Step 1**: find the mean and covariance of $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$

$$p(Z_0) = \mathcal{N}(\boldsymbol{\pi}_0, \Sigma_0)$$
$$p(X_t \mid \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

Gaussian Combining: If $\mathbf{x} = \mathbf{C}\mathbf{z} + \mathbf{v}$ with $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \Sigma_z)$ and $\mathbf{v} \sim \mathcal{N}(0, \Sigma_v)$ then

$$\begin{bmatrix} \mathbf{z} \\ \mathbf{x} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_z \\ \mathbf{C}\boldsymbol{\mu}_z \end{bmatrix}, \begin{bmatrix} \Sigma_z & \Sigma_z \mathbf{C}^\top \\ \mathbf{C}\Sigma_z & \mathbf{C}\Sigma_z \mathbf{C}^\top + \Sigma_v \end{bmatrix} \right)$$

- Applying **Gaussian combining** yields

$$\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\pi}_0 \\ \mathbf{C}\boldsymbol{\pi}_0 \end{bmatrix}, \begin{bmatrix} \Sigma_0 & \Sigma_0 \mathbf{C}^\top \\ \mathbf{C}\Sigma_0 & \mathbf{C}\Sigma_0 \mathbf{C}^\top + \mathbf{R} \end{bmatrix} \right)$$

Filtering: Compute $\hat{\mathbf{z}}_{0|0}, \hat{\Sigma}_{0|0}$

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

- Step 2: apply Gaussian conditioning to $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{b} - \boldsymbol{\mu}_b), \quad \Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

$$\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\pi}_0 \\ \mathbf{C}\boldsymbol{\pi}_0 \end{bmatrix}, \begin{bmatrix} \Sigma_0 & \Sigma_0\mathbf{C}^\top \\ \mathbf{C}\Sigma_0 & \mathbf{C}\Sigma_0\mathbf{C}^\top + \mathbf{R} \end{bmatrix}\right)$$

- We have

$$\hat{\mathbf{z}}_{0|0} = \mathbb{E}[\mathbf{z}_0 \mid \mathbf{x}_0] = \boldsymbol{\pi}_0 + \Sigma_0\mathbf{C}^\top(\mathbf{C}\Sigma_0\mathbf{C}^\top + \mathbf{R})^{-1}(\mathbf{x}_0 - \mathbf{C}\boldsymbol{\pi}_0)$$

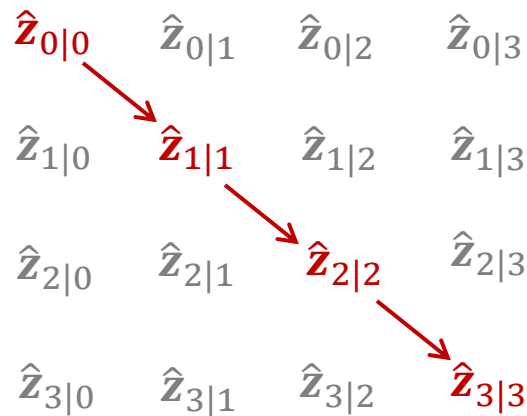
$$\hat{\Sigma}_{0|0} = \text{Cov}(\mathbf{z}_0 \mid \mathbf{x}_0) = \Sigma_0 - \Sigma_0\mathbf{C}^\top(\mathbf{C}\Sigma_0\mathbf{C}^\top + \mathbf{R})^{-1}\mathbf{C}\Sigma_0$$

Filtering: From 0 to t

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **P1: Filtering.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_t)$, compute $p_\theta(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$
- We've now computed $\hat{\mathbf{z}}_{0|0}$ and $\hat{\Sigma}_{0|0}$. This solves **P1** for the case $t = 0$
- To proceed, we will update $\hat{\mathbf{z}}_{t-1|t-1}, \hat{\Sigma}_{t-1|t-1}$ into $\hat{\mathbf{z}}_{t|t}, \hat{\Sigma}_{t|t}$ for every t

The planned computational flow



Filtering: From 0 to t

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- To compute $\hat{\mathbf{z}}_{0|0}$ and $\hat{\Sigma}_{0|0}$, we
 - (Step 0) found the mean and covariance of $\mathbf{z}_0$ (already known)
 - (Step 1) found the mean and covariance of $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$ via **Gaussian combining**
 - (Step 2) found the mean and covariance of $\mathbf{z}_0 \mid \mathbf{x}_0$ via **Gaussian conditioning**

Question: How can we generalize these steps for general t ?

- To update $\hat{\mathbf{z}}_{t-1|t-1}$, $\hat{\Sigma}_{t-1|t-1}$ into $\hat{\mathbf{z}}_{t|t}$, $\hat{\Sigma}_{t|t}$, we will **condition on $\mathbf{x}_0, \dots, \mathbf{x}_{t-1}$** and
 - (Step 0) find the mean and covariance of $\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ (using $\hat{\mathbf{z}}_{t-1|t-1}$, $\hat{\Sigma}_{t-1|t-1}$)
 - (Step 1) find the mean and covariance of $\begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ via **Gaussian combining**
 - (Step 2) find the mean and covariance of $\mathbf{z}_t \mid \mathbf{x}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ via **Gaussian conditioning**

Step 0 and conditioning on $\mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ are the only differences

- **You should be able to figure out all the details without looking at the rest slides**

Filtering: Compute $\hat{\mathbf{z}}_{t|t}, \hat{\Sigma}_{t|t}$

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

- **Step 0:** find the mean and covariance of $\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$
 - By definition, this is to compute $\hat{\mathbf{z}}_{t|t-1}, \hat{\Sigma}_{t|t-1}$

$$p(\mathbf{z}_t \mid \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

- We have

$$\hat{\mathbf{z}}_{t|t-1} = \mathbb{E}[\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}] = \mathbb{E}[\mathbf{A}\mathbf{z}_{t-1} + \mathbf{w}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}] = \mathbf{A}\hat{\mathbf{z}}_{t-1|t-1}$$

$$\hat{\Sigma}_{t|t-1} = \mathbb{E} \left[(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1})(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1})^\top \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1} \right] = \dots = \mathbf{A}\hat{\Sigma}_{t-1|t-1}\mathbf{A}^\top + \mathbf{Q}$$

↑
similar to how we computed $\text{Cov}(\mathbf{x})$ in
the proof of **Gaussian combining**

Filtering: Compute $\hat{\mathbf{z}}_{t|t}, \hat{\Sigma}_{t|t}$

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **Step 1**: find the mean and covariance of $\begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$

$$\hat{\mathbf{z}}_{t|t-1} = A\hat{\mathbf{z}}_{t-1|t-1}$$
$$\hat{\Sigma}_{t|t-1} = A\hat{\Sigma}_{t-1|t-1}A^\top + Q$$

$$\hat{\mathbf{z}}_{0|-1} := \boldsymbol{\pi}_0, \hat{\Sigma}_{0|-1} := \Sigma_0$$

- We applied **Gaussian combining** to $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$ and obtained

$$\bullet \begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mathbf{z}}_{0|-1} \\ C\hat{\mathbf{z}}_{0|-1} \end{bmatrix}, \begin{bmatrix} \hat{\Sigma}_{0|-1} & \hat{\Sigma}_{0|-1}C^\top \\ C\hat{\Sigma}_{0|-1} & C\hat{\Sigma}_{0|-1}C^\top + R \end{bmatrix} \right)$$

- Similarly, now, applying **Gaussian combining** to $\begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ gives:

$$\bullet \begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1} \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mathbf{z}}_{t|t-1} \\ C\hat{\mathbf{z}}_{t|t-1} \end{bmatrix}, \begin{bmatrix} \hat{\Sigma}_{t|t-1} & \hat{\Sigma}_{t|t-1}C^\top \\ C\hat{\Sigma}_{t|t-1} & C\hat{\Sigma}_{t|t-1}C^\top + R \end{bmatrix} \right)$$

Filtering: Compute $\hat{\mathbf{z}}_{t|t}, \hat{\Sigma}_{t|t}$

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

- Step 2: apply Gaussian conditioning to $\begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$

$$\begin{aligned}\hat{\mathbf{z}}_{t|t-1} &= A\hat{\mathbf{z}}_{t-1|t-1} \\ \hat{\Sigma}_{t|t-1} &= A\hat{\Sigma}_{t-1|t-1}A^\top + Q\end{aligned}$$

$$\hat{\mathbf{z}}_{0|-1} := \boldsymbol{\pi}_0, \quad \hat{\Sigma}_{0|-1} := \Sigma_0$$

- We applied Gaussian conditioning to $\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{x}_0 \end{bmatrix}$ and obtained:

$$\hat{\mathbf{z}}_{0|0} = \hat{\mathbf{z}}_{0|-1} + \hat{\Sigma}_{0|-1} \mathbf{C}^\top (\mathbf{C} \hat{\Sigma}_{0|-1} \mathbf{C}^\top + \mathbf{R})^{-1} (\mathbf{x}_0 - \mathbf{C} \hat{\mathbf{z}}_{0|-1})$$

$$\hat{\Sigma}_{0|0} = \hat{\Sigma}_{0|-1} - \hat{\Sigma}_{0|-1} \mathbf{C}^\top (\mathbf{C} \hat{\Sigma}_{0|-1} \mathbf{C}^\top + \mathbf{R})^{-1} \mathbf{C} \hat{\Sigma}_{0|-1}$$

- Similarly, now, applying Gaussian conditioning to $\begin{bmatrix} \mathbf{z}_t \\ \mathbf{x}_t \end{bmatrix} \mid \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ gives:

$$\hat{\mathbf{z}}_{t|t} = \hat{\mathbf{z}}_{t|t-1} + \hat{\Sigma}_{t|t-1} \mathbf{C}^\top (\mathbf{C} \hat{\Sigma}_{t|t-1} \mathbf{C}^\top + \mathbf{R})^{-1} (\mathbf{x}_t - \mathbf{C} \hat{\mathbf{z}}_{t|t-1})$$

$$\hat{\Sigma}_{t|t} = \hat{\Sigma}_{t|t-1} - \underbrace{\hat{\Sigma}_{t|t-1} \mathbf{C}^\top (\mathbf{C} \hat{\Sigma}_{t|t-1} \mathbf{C}^\top + \mathbf{R})^{-1}}_{\text{Kalman gain matrix}} \mathbf{C} \hat{\Sigma}_{t|t-1}$$

“Kalman gain matrix”. Let us denote it by $\mathbf{K}_t$

Summary: Filtering for LDS

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

$$p(\mathbf{Z}_0) = \mathcal{N}(\boldsymbol{\pi}_0, \Sigma_0)$$
$$p(\mathbf{Z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$
$$p(\mathbf{X}_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

- Putting everything together gives **Kalman Filter**:

- Initialization: $\hat{\mathbf{z}}_{0|-1} := \boldsymbol{\pi}_0$, $\hat{\Sigma}_{0|-1} := \Sigma_0$

- Recursion ($\forall t = 0, \dots, T$):

“Correction”:

$$\mathbf{K}_t = \hat{\Sigma}_{t|t-1} \mathbf{C}^\top (\mathbf{C} \hat{\Sigma}_{t|t-1} \mathbf{C}^\top + \mathbf{R})^{-1}$$

$$\hat{\mathbf{z}}_{t|t} = \hat{\mathbf{z}}_{t|t-1} + \mathbf{K}_t (\mathbf{x}_t - \mathbf{C} \hat{\mathbf{z}}_{t|t-1})$$

$$\hat{\Sigma}_{t|t} = \hat{\Sigma}_{t|t-1} - \mathbf{K}_t \mathbf{C} \hat{\Sigma}_{t|t-1}$$

“Prediction”:

$$\hat{\mathbf{z}}_{t+1|t} = \mathbf{A} \hat{\mathbf{z}}_{t|t}$$

$$\hat{\Sigma}_{t+1|t} = \mathbf{A} \hat{\Sigma}_{t|t} \mathbf{A}^\top + \mathbf{Q}$$

How To Derive Kalman Filter

$$p(\mathbf{z}_{t-1} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$$

Prediction

$$p(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$$

Gaussian Combining

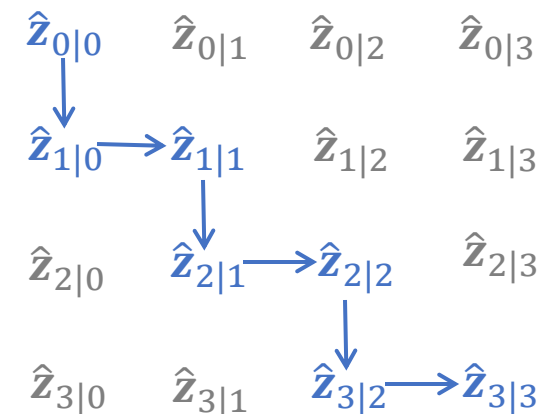
$$p(\mathbf{z}_t, \mathbf{x}_t | \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$$

Gaussian Conditioning

$$p(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_{t-1}, \mathbf{x}_t)$$

Correction

Computational Trajectory of Kalman Filter



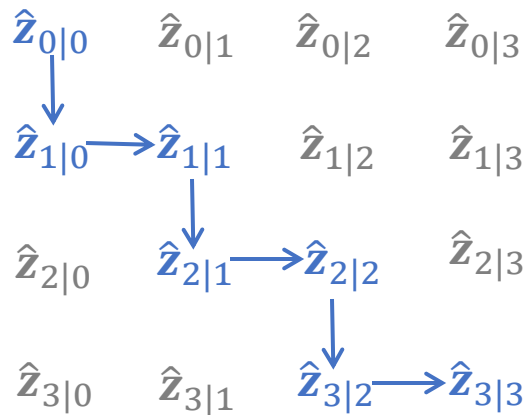
From Filtering to Smoothing

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$

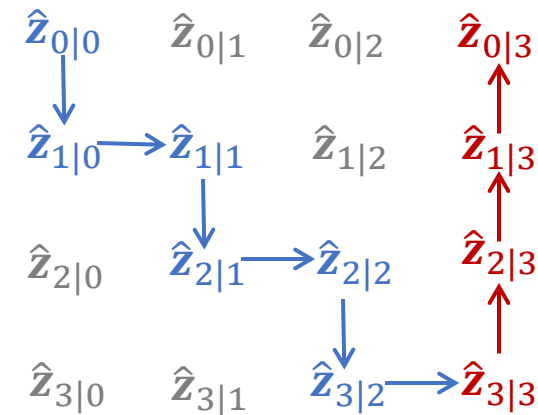
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **P1: Filtering**. Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_t)$, compute $p_\theta(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_t)$
- **P2: Smoothing**. Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, compute $p_\theta(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_T)$
- Since everything is Gaussian, to solve **P2** it suffices to compute $\hat{\mathbf{z}}_{t|T}, \hat{\Sigma}_{t|T}$ for all t

What **Kalman Filter** gives us:



What we will do to solve **P2**:



Goal: Given $\hat{\mathbf{z}}_{t|t}, \hat{\Sigma}_{t|t}$ and $\hat{\mathbf{z}}_{t|t-1}, \hat{\Sigma}_{t|t-1}$ for every t , update $\hat{\mathbf{z}}_{t|T}, \hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}, \hat{\Sigma}_{t-1|T}$

Smoothing

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

Goal: Given $\hat{\mathbf{z}}_{t|t}$, $\hat{\Sigma}_{t|t}$ and $\hat{\mathbf{z}}_{t|t-1}$, $\hat{\Sigma}_{t|t-1}$ for every t , update $\hat{\mathbf{z}}_{t|T}$, $\hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}$, $\hat{\Sigma}_{t-1|T}$

- **Observation:**

- Since $\mathbf{z}_{t-1}$ is independent of $\mathbf{x}_t, \dots, \mathbf{x}_T$ given $\mathbf{z}_t$, we have

$$p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1}) = p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$$

- **High-level Idea:**

1. Compute $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$ via **Gaussian combining** and **Gaussian conditioning**
 - This gives us $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$
2. Given $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$, update $\hat{\mathbf{z}}_{t|T}$, $\hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}$, $\hat{\Sigma}_{t-1|T}$
 - via *law of total probability* and *law of total expectation*

- **Derivation Details:** see the appendix at the end of slides

Summary: Smoothing for LDSs

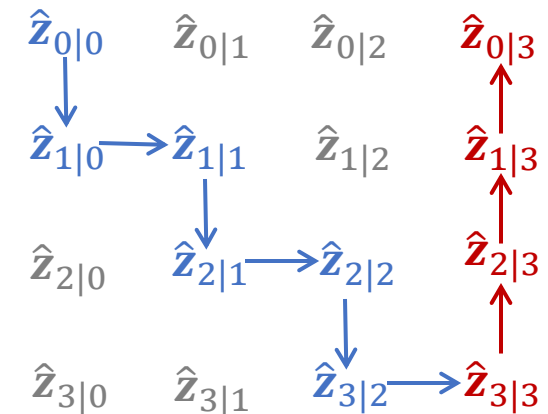
$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$

$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **Goal.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, compute

$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_T)$$
- **Algorithm** (known as “Rauch-Tung-Striebel smoother”).
 1. (Forward Pass) Run Kalman filtering to compute $\hat{\mathbf{z}}_{t|t}$, $\hat{\Sigma}_{t|t}$ and $\hat{\mathbf{z}}_{t+1|t}$, $\hat{\Sigma}_{t+1|t}$ for all t
 2. (Backward Pass) For $t = T, \dots, 1$, compute the following:
 - $\mathbf{L}_{t-1} = \hat{\Sigma}_{t-1|t-1} \mathbf{A}^{\top} \hat{\Sigma}_{t|t-1}^{-1}$
 - $\hat{\mathbf{z}}_{t-1|T} = \hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1} (\hat{\mathbf{z}}_{t|T} - \hat{\mathbf{z}}_{t|t-1})$
 - $\hat{\Sigma}_{t-1|T} = \hat{\Sigma}_{t-1|t-1} + \mathbf{L}_{t-1} (\hat{\Sigma}_{t|T} - \hat{\Sigma}_{t|t-1}) \mathbf{L}_{t-1}^{\top}$

Computational Trajectory of Smoothing:



Remark: $\mathbf{L}_{t-1}$ is sometimes called the “backward Kalman gain matrix”

State Estimation and Learning

- State Estimation (“Decoding”). Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, solve:

$$\operatorname{argmax}_{\mathbf{z}_0, \dots, \mathbf{z}_T} p_{\theta}(\mathbf{z}_0, \dots, \mathbf{z}_T \mid \mathbf{x}_0, \dots, \mathbf{x}_T)$$

- Learning. Given N observations $\{\mathbf{x}_{0:T}^{(n)}\}_{n=1}^N$, find best θ :

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(\mathbf{x}_{0:T}^{(n)})$$

State Estimation

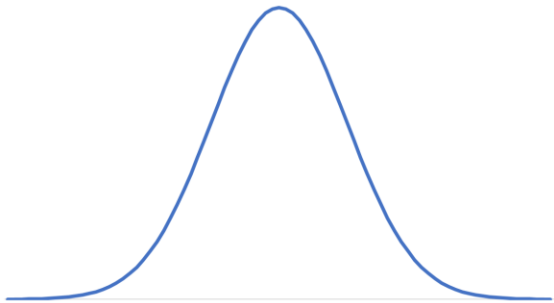
$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **State Estimation.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, solve:

$$\underset{\mathbf{z}_0, \dots, \mathbf{z}_T}{\operatorname{argmax}} p_{\theta}(\mathbf{z}_0, \dots, \mathbf{z}_T | \mathbf{x}_0, \dots, \mathbf{x}_T)$$

- **Solution:**

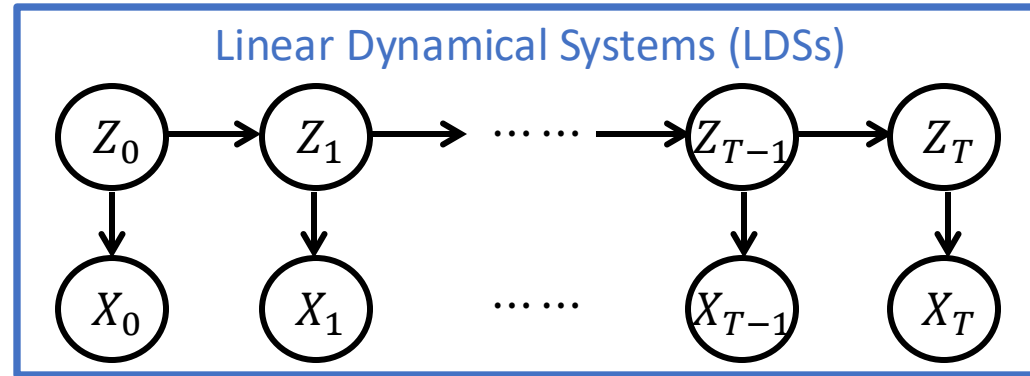
- Since $p_{\theta}(\mathbf{z}_0, \dots, \mathbf{z}_T | \mathbf{x}_0, \dots, \mathbf{x}_T)$ is Gaussian, the optimal solution to state estimation is



$$\mathbb{E} \left[\begin{bmatrix} \mathbf{z}_0 \\ \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_T \end{bmatrix} \middle| \mathbf{x}_0, \dots, \mathbf{x}_T \right] = \begin{bmatrix} \mathbb{E}[\mathbf{z}_0 | \mathbf{x}_0, \dots, \mathbf{x}_T] \\ \mathbb{E}[\mathbf{z}_1 | \mathbf{x}_0, \dots, \mathbf{x}_T] \\ \vdots \\ \mathbb{E}[\mathbf{z}_T | \mathbf{x}_0, \dots, \mathbf{x}_T] \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{z}}_{0|T} \\ \hat{\mathbf{z}}_{1|T} \\ \vdots \\ \hat{\mathbf{z}}_{T|T} \end{bmatrix}$$

- Therefore, the state estimation problem can be solved by smoothing
- Similarly, we can prove the output $\hat{\mathbf{z}}_{t|t}$ of the Kalman filter solves $\max_{\mathbf{z}_t} p_{\theta}(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_t)$. It is in this sense that we say the Kalman filter is optimal

Learning



Model Parameters:

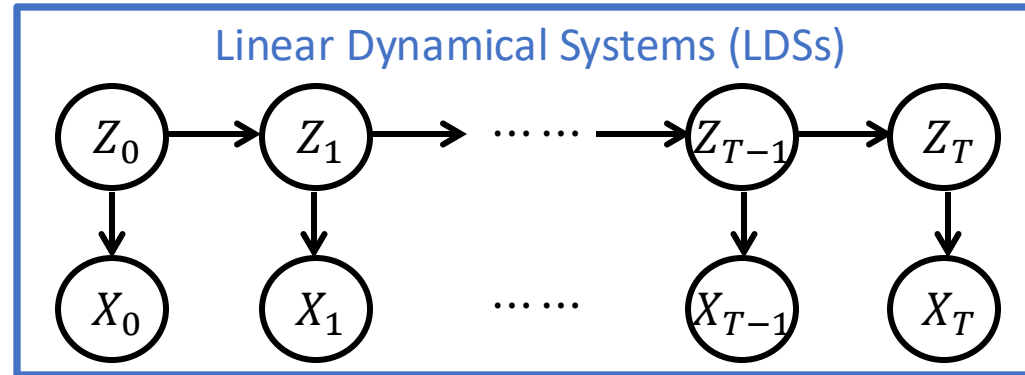
- $\theta := (\pi_0, \Sigma_0, A, Q, C, R)$

- **Learning.** Given N observations $\{\mathbf{x}_{0:T}^{(n)}\}_{n=1}^N$, find best θ :

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(\mathbf{x}_{0:T}^{(n)})$$

- **We can perform learning via the EM algorithm**
 - we provide high level intuition in what follows
 - full derivations are given at the end of the slides

Learning



Model Parameters:

- $\theta := (\pi_0, \Sigma_0, A, Q, C, R)$

- **Learning.** Given N observations $\{\mathbf{x}_{0:T}^{(n)}\}_{n=1}^N$, find best θ :

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(\mathbf{x}_{0:T}^{(n)})$$

- We are going to apply the EM algorithm (iteration: k):

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} \left[\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T}) \right]$$

- we provide high level intuition in what follows
- full derivations are given at the end of the slides

in subscripts: $\mathbf{z} := \mathbf{z}_{0:T}, \mathbf{x}^{(n)} := \mathbf{x}_{0:T}^{(n)}$

Guessing

Model Parameters:

- $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\log p_{\theta}(\mathbf{x}_{0:T}^{(n)} | \mathbf{z}_{0:T})]$$

$$p(Z_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

$$p(Z_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

$$p(Y_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

- Let us guess a solution to the M-step... (one of the guesses is wrong)

- Since $\boldsymbol{\pi}_0 = \mathbb{E}[\mathbf{z}_0]$, we guess:

$$\boldsymbol{\pi}_0^{k+1} = \sum_{n=1}^N \frac{\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0]}{N} \quad \leftarrow \text{“empirical mean”}$$

- And we guess the covariance might be

$$\boldsymbol{\Sigma}_0^{k+1} = \sum_{n=1}^N \frac{\left(\mathbb{E}_{q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0] - \boldsymbol{\pi}_0^{k+1} \right) \left(\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0] - \boldsymbol{\pi}_0^{k+1} \right)^{\top}}{N} \quad \leftarrow \text{“empirical covariance”}$$

- Try having a guess for $\boldsymbol{\Sigma}_0^{k+1}, \mathbf{A}^{k+1}, \mathbf{Q}^{k+1}, \mathbf{C}^{k+1}, \mathbf{R}^{k+1}$ yourself...

Summary: EM for LDSs (iteration: k)

E-step

Given θ^k , for each $\mathbf{x}_{0:T}^{(n)}$, use **Kalman Filter** & **Smoothing** to compute:

- $\mathbb{E}_k^{(n)}[\mathbf{z}_t] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t | \mathbf{x}_{0:T}^{(n)}]$
- $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t \mathbf{z}_t^\top | \mathbf{x}_{0:T}^{(n)}]$
- $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t \mathbf{z}_{t-1}^\top | \mathbf{x}_{0:T}^{(n)}]$

← add indices n, k

M-step

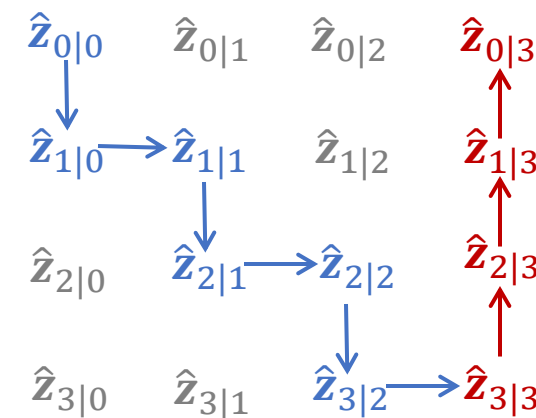
Update parameters:

- $\boldsymbol{\pi}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0]}{N}$
- $\boldsymbol{\Sigma}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top]}{N} - \boldsymbol{\pi}_0^{k+1} (\boldsymbol{\pi}_0^{k+1})^\top$
- $\mathbf{C}^{k+1} = \left(\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} \mathbb{E}_k^{(n)}[\mathbf{z}_t]^\top \right) \left(\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \right)^{-1}$
- $\mathbf{R}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} (\mathbf{x}_t^{(n)})^\top - \mathbf{C}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t] (\mathbf{x}_t^{(n)})^\top}{N(T+1)}$
- $\mathbf{A}^{k+1} = \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] \right) \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top] \right)^{-1}$
- $\mathbf{Q}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] - \mathbf{A}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_t^\top]}{NT}$

- $\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$
- $\hat{\boldsymbol{\Sigma}}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$
- $\mathbf{L}_{t-1} := \hat{\boldsymbol{\Sigma}}_{t-1|t-1} \mathbf{A}^\top \hat{\boldsymbol{\Sigma}}_{t|t-1}^{-1}$

Kalman Filter & **Smoothing** can compute

- $\mathbb{E}[\mathbf{z}_t | \mathbf{x}_{0:T}] = \hat{\mathbf{z}}_{t|T}$
 - $\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^\top | \mathbf{x}_{0:T}] = \hat{\boldsymbol{\Sigma}}_{t|T} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t|T}^\top$
 - $\mathbb{E}[\mathbf{z}_t \mathbf{z}_{t-1}^\top | \mathbf{x}_{0:T}] = \mathbf{L}_{t-1} \hat{\boldsymbol{\Sigma}}_{t|T} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t-1|T}^\top$
- via **forward** & **backward** passes



Model Parameters:

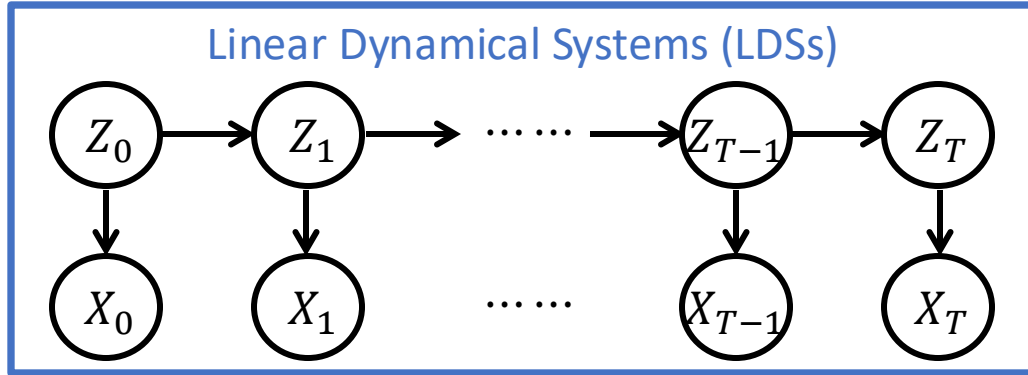
- $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

$$p(\mathbf{Z}_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

$$p(\mathbf{Z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A} \mathbf{z}_{t-1}, \mathbf{Q})$$

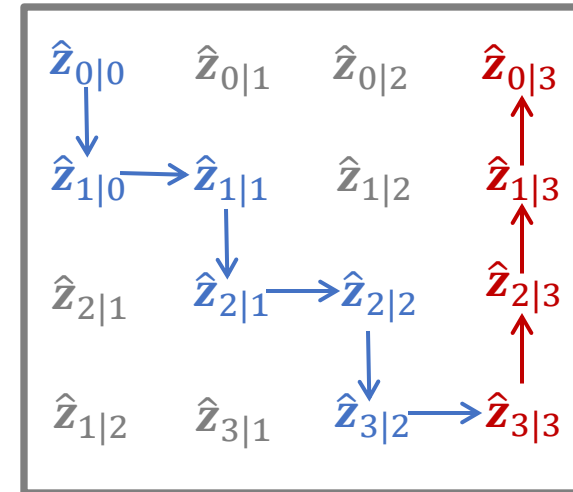
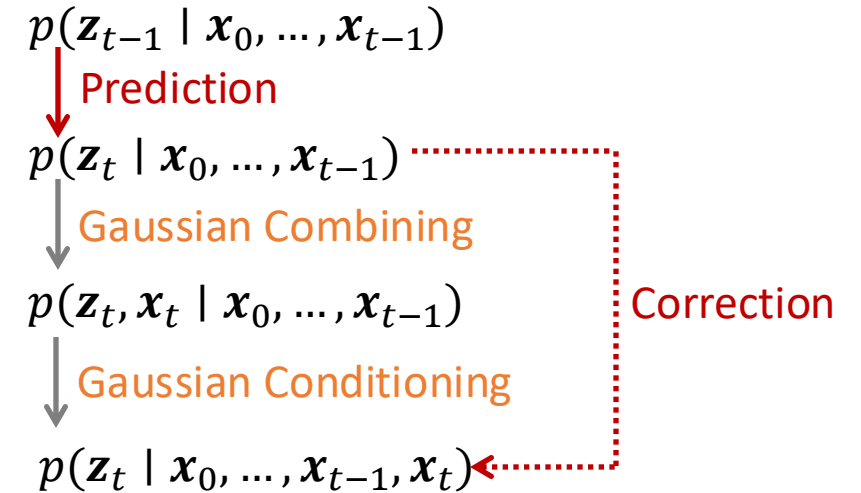
$$p(\mathbf{X}_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C} \mathbf{z}_t, \mathbf{R})$$

Possible Extensions of Linear Dynamical Systems



- What if
 - we do **not** have Gaussians?
 - we have time-varying dynamics?
 - $Z_t = \mathbf{A}_t \mathbf{z}_{t-1} + \mathbf{w}_t$, $X_t = \mathbf{C}_t \mathbf{z}_t + \mathbf{v}_t$
 - we have **control** over states?
 - $Z_t = \mathbf{A} \mathbf{z}_{t-1} + \mathbf{U} \mathbf{q}_t + \mathbf{w}_t$
 - we have **nonlinear** dynamics?
 - $Z_t = f(\mathbf{z}_{t-1}) + \mathbf{w}_t$, $X_t = g(\mathbf{z}_t) + \mathbf{v}_t$

How To Derive Kalman Filter



Appendix (Optional)

- In the subsequent slides, we provide mathematical details for smoothing and learning

Smoothing: Prerequisite

- For smoothing, we need the following basic results:
 - Law of Total Expectation (LoTE)

$$\mathbb{E}[\mathbf{a}] = \mathbb{E}_b[\mathbb{E}_a[\mathbf{a} \mid \mathbf{b}]]$$

- Law of Total Covariance (LoTC)

$$\text{Cov}(\mathbf{a}) = \mathbb{E}[\text{Cov}(\mathbf{a} \mid \mathbf{b})] + \text{Cov}(\mathbb{E}[\mathbf{a} \mid \mathbf{b}])$$

$$\begin{aligned}\text{Cov}(\mathbf{a}, \mathbf{b}) &:= \mathbb{E}[(\mathbf{a} - \mathbb{E}[\mathbf{a}])(\mathbf{b} - \mathbb{E}[\mathbf{b}])^\top] \\ \text{Cov}(\mathbf{a}) &:= \text{Cov}(\mathbf{a}, \mathbf{a})\end{aligned}$$

Smoothing

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

Goal: Given $\hat{\mathbf{z}}_{t|t}$, $\hat{\Sigma}_{t|t}$ and $\hat{\mathbf{z}}_{t|t-1}$, $\hat{\Sigma}_{t|t-1}$ for every t , update $\hat{\mathbf{z}}_{t|T}$, $\hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}$, $\hat{\Sigma}_{t-1|T}$

- **Observation:**

- Since $\mathbf{z}_{t-1}$ is independent of $\mathbf{x}_t, \dots, \mathbf{x}_T$ given $\mathbf{z}_t$, we have

$$p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1}) = p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$$

- **High-level Idea:**

1. Compute $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$ via **Gaussian combining** and **Gaussian conditioning**
 - This gives us $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$
2. Given $p_{\theta}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)$, update $\hat{\mathbf{z}}_{t|T}$, $\hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}$, $\hat{\Sigma}_{t-1|T}$ via **LoTE** and **LoTC**

Smoothing

$$p(Z_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$

$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **Step 1:** Compute $p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1})$

Gaussian Combining: If $\mathbf{x} = \mathbf{C}\mathbf{z} + \mathbf{v}$ with $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \Sigma_z)$ and $\mathbf{v} \sim \mathcal{N}(0, \Sigma_v)$ then

$$\begin{bmatrix} \mathbf{z} \\ \mathbf{x} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_z \\ \mathbf{C}\boldsymbol{\mu}_z \end{bmatrix}, \begin{bmatrix} \Sigma_z & \Sigma_z \mathbf{C}^\top \\ \mathbf{C}\Sigma_z & \mathbf{C}\Sigma_z \mathbf{C}^\top + \Sigma_v \end{bmatrix} \right)$$

Gaussian Conditioning: $\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{b} - \boldsymbol{\mu}_b)$, $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$

- **Step 1.1:** Applying **Gaussian combining** to $\begin{bmatrix} \mathbf{z}_{t-1} \\ \mathbf{z}_t \end{bmatrix} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ gives:

$$\begin{bmatrix} \mathbf{z}_{t-1} \\ \mathbf{z}_t \end{bmatrix} | \mathbf{x}_0, \dots, \mathbf{x}_{t-1} \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mathbf{z}}_{t-1|t-1} \\ \hat{\mathbf{z}}_{t|t-1} \end{bmatrix}, \begin{bmatrix} \hat{\Sigma}_{t-1|t-1} & \hat{\Sigma}_{t-1|t-1} \mathbf{A}^\top \\ \mathbf{A} \hat{\Sigma}_{t-1|t-1} & \hat{\Sigma}_{t|t-1} \end{bmatrix} \right)$$

$$\hat{\Sigma}_{t|t-1} = \mathbf{A} \hat{\Sigma}_{t-1|t-1} \mathbf{A}^\top + \mathbf{Q}$$

- **Step 1.2:** From **Gaussian conditioning** we see $\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_{t-1}$ has distribution:

$$\mathcal{N}(\hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1}), \hat{\Sigma}_{t-1|t-1} - \mathbf{L}_{t-1} \mathbf{A} \hat{\Sigma}_{t-1|t-1})$$

$$\mathbf{L}_{t-1} := \hat{\Sigma}_{t-1|t-1} \mathbf{A}^\top \hat{\Sigma}_{t|t-1}^{-1}$$

Remark. Note that $\mathbf{L}_{t-1} \mathbf{A} \hat{\Sigma}_{t-1|t-1} = \mathbf{L}_{t-1} \hat{\Sigma}_{t|t-1} \mathbf{L}_{t-1}^\top$, so the covariance $\hat{\Sigma}_{t-1|t-1} - \mathbf{L}_{t-1} \mathbf{A} \hat{\Sigma}_{t-1|t-1}$ is symmetric

Smoothing

$$\mathbf{L}_{t-1} = \hat{\Sigma}_{t-1|t-1} \mathbf{A}^\top \hat{\Sigma}_{t|t-1}^{-1}$$

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

- We have obtained

$$p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T) = \mathcal{N}(\hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1}), \hat{\Sigma}_{t-1|t-1} - \mathbf{L}_{t-1} \hat{\Sigma}_{t|t-1} \mathbf{L}_{t-1}^\top)$$

- Step 2: update $\hat{\mathbf{z}}_{t|T}, \hat{\Sigma}_{t|T}$ into $\hat{\mathbf{z}}_{t-1|T}, \hat{\Sigma}_{t-1|T}$ via LoTE and LoTC

$$\begin{aligned}\hat{\mathbf{z}}_{t-1|T} &= \mathbb{E}[\mathbf{z}_{t-1} \mid \mathbf{x}_0, \dots, \mathbf{x}_T] = \mathbb{E}_{\mathbf{z}_t}[\mathbb{E}_{\mathbf{z}_{t-1}}[\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T]] \\ &= \mathbb{E}_{\mathbf{z}_t}[\hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1}) \mid \mathbf{x}_0, \dots, \mathbf{x}_T] \\ &= \hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\hat{\mathbf{z}}_{t|T} - \hat{\mathbf{z}}_{t|t-1})\end{aligned}$$

$$\begin{aligned}\hat{\Sigma}_{t-1|T} &= \text{Cov}(\mathbf{z}_{t-1} \mid \mathbf{x}_0, \dots, \mathbf{x}_T) = \mathbb{E}[\text{Cov}(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T)] + \text{Cov}(\mathbb{E}[\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0, \dots, \mathbf{x}_T]) \\ &= \mathbb{E}[\hat{\Sigma}_{t-1|t-1} - \mathbf{L}_{t-1} \hat{\Sigma}_{t|t-1} \mathbf{L}_{t-1}^\top] + \text{Cov}(\hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1})) \\ &= \hat{\Sigma}_{t-1|t-1} - \mathbf{L}_{t-1} \hat{\Sigma}_{t|t-1} \mathbf{L}_{t-1}^\top + \text{Cov}(\hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1}(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1}) \mid \mathbf{x}_0, \dots, \mathbf{x}_T) \\ &= \hat{\Sigma}_{t-1|t-1} + \mathbf{L}_{t-1}(\hat{\Sigma}_{t|T} - \hat{\Sigma}_{t|t-1}) \mathbf{L}_{t-1}^\top\end{aligned}$$

Summary: Smoothing for LDSs

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t]$$

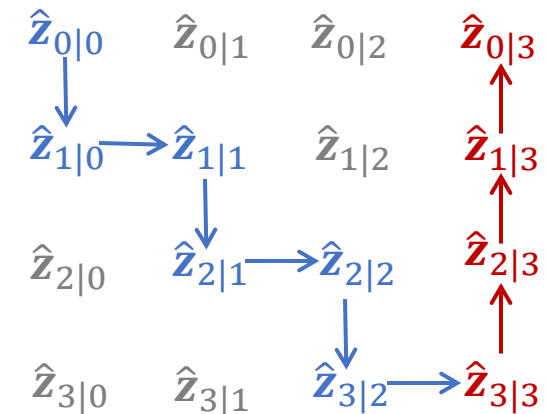
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s \mid \mathbf{x}_0, \dots, \mathbf{x}_t)$$

- **Goal.** Given θ and $(\mathbf{x}_0, \dots, \mathbf{x}_T)$, compute

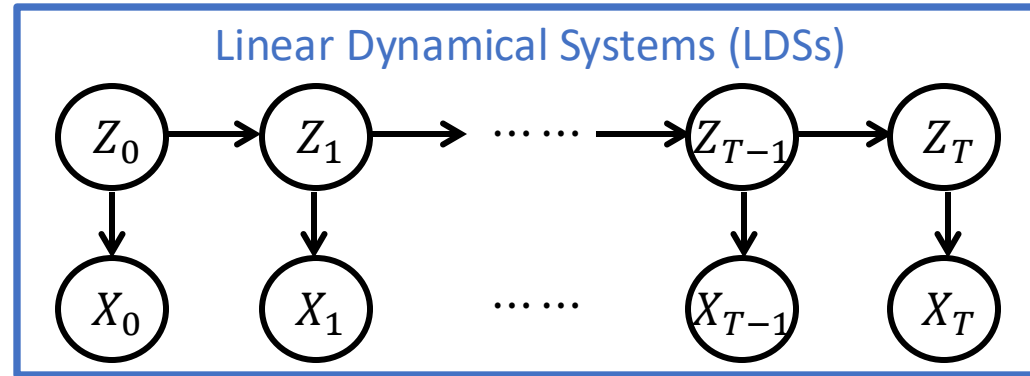
$$p_{\theta}(\mathbf{z}_t \mid \mathbf{x}_0, \dots, \mathbf{x}_T)$$
- **Algorithm** (known as “Rauch-Tung-Striebel smoother”).
 1. (Forward Pass) Run Kalman filtering to compute $\hat{\mathbf{z}}_{t|t}$, $\hat{\Sigma}_{t|t}$ and $\hat{\mathbf{z}}_{t+1|t}$, $\hat{\Sigma}_{t+1|t}$ for all t
 2. (Backward Pass) For $t = T, \dots, 1$, compute the following:
 - $\mathbf{L}_{t-1} = \hat{\Sigma}_{t-1|t-1} \mathbf{A}^T \hat{\Sigma}_{t|t-1}^{-1}$
 - $\hat{\mathbf{z}}_{t-1|T} = \hat{\mathbf{z}}_{t-1|t-1} + \mathbf{L}_{t-1} (\hat{\mathbf{z}}_{t|T} - \hat{\mathbf{z}}_{t|t-1})$
 - $\hat{\Sigma}_{t-1|T} = \hat{\Sigma}_{t-1|t-1} + \mathbf{L}_{t-1} (\hat{\Sigma}_{t|T} - \hat{\Sigma}_{t|t-1}) \mathbf{L}_{t-1}^T$

Remark: $\mathbf{L}_{t-1}$ might also be computed in the forward pass

Computational Trajectory of Smoothing:



Learning



Model Parameters:

- $\theta := (\pi_0, \Sigma_0, A, Q, C, R)$

- **Learning.** Given N observations $\{\mathbf{x}_{0:T}^{(n)}\}_{n=1}^N$, find best θ :

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(\mathbf{x}_{0:T}^{(n)})$$

- **We can perform learning via the EM algorithm**
 - derivations are algebraically involved and are given at the end of the slides.

Learning (Review Lecture 2)

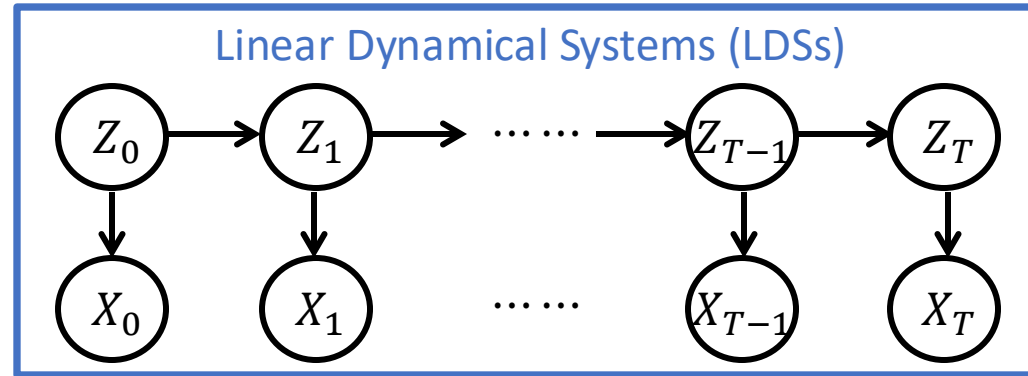
- The likelihood $\prod_{i=1}^N p_{\theta}(\mathbf{x}_i) = \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})\right)}{(2\pi)^{\frac{ND}{2}} \det(\boldsymbol{\Sigma})^{\frac{N}{2}}}$ is maximized at

$$\boldsymbol{\mu}^* = \frac{\sum_{i=1}^N \mathbf{x}_i}{N}, \quad \boldsymbol{\Sigma}^* = \frac{\sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu}^*)(\mathbf{x}_i - \boldsymbol{\mu}^*)^{\top}}{N}$$

← empirical mean ← empirical covariance

- How did we obtain $\boldsymbol{\mu}^*$ and $\boldsymbol{\Sigma}^*$?
 - **Step 0:** Rewrite the objective
 - maximizing log-likelihood is minimizing $N \log \det \boldsymbol{\Sigma} + \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})$
 - **Step 1:** Set the derivative w.r.t. $\boldsymbol{\mu}$ to 0
 - solving it gives $\boldsymbol{\mu}^*$
 - **Step 2:** Substitute $\boldsymbol{\mu} = \boldsymbol{\mu}^*$ into the objective, and set the derivative w.r.t. $\boldsymbol{\Sigma}^{-1}$ to 0
 $m \cdot \log \det \boldsymbol{\Sigma} + \text{tr}(\mathbf{S}\boldsymbol{\Sigma}^{-1})$ is minimized at $\boldsymbol{\Sigma} = \mathbf{S}/m$

Learning



Model Parameters:

- $\theta := (\pi_0, \Sigma_0, A, Q, C, R)$

- **Learning.** Given N observations $\{\mathbf{x}_{0:T}^{(n)}\}_{n=1}^N$, find best θ :

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(\mathbf{x}_{0:T}^{(n)})$$

- We are going to apply the EM algorithm (iteration: k):

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} \left[\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T}) \right]$$

Guessing

Model Parameters:

- $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\log p_{\theta}(\mathbf{x}_{0:T}^{(n)} | \mathbf{z}_{0:T})]$$

$$p(Z_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

$$p(Z_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

$$p(Y_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

- Let guess a solution to the M-step:

- Since $\boldsymbol{\pi}_0 = \mathbb{E}[\mathbf{z}_0]$, we guess... :

$$\boldsymbol{\pi}_0^{k+1} = \sum_{n=1}^N \frac{\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0]}{N} \quad \leftarrow \text{“empirical mean”}$$

- And we guess the covariance should be

$$\boldsymbol{\Sigma}_0^{k+1} = \sum_{n=1}^N \frac{\left(\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0] - \boldsymbol{\pi}_0^{k+1} \right) \left(\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\mathbf{z}_0] - \boldsymbol{\pi}_0^{k+1} \right)^{\top}}{N} \quad \leftarrow \text{“empirical covariance”}$$

- Try having a guess for $\mathbf{A}^{k+1}, \mathbf{Q}^{k+1}, \mathbf{C}^{k+1}, \mathbf{R}^{k+1}$ yourself...

E-step (iteration: k)

$$\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$$
$$\hat{\Sigma}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$$

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T})]$$

- In E-step, we will need to compute the expectation $\mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\cdot]$.
- “It turns out that” we only need to compute the following expectations:

$$\mathbb{E}_k^{(n)}[\mathbf{z}_t] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t]$$
$$\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^{\top}] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_t^{\top}]$$
$$\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^{\top}] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^{\top}]$$

- later you will see why...

E-step (iteration: k)

$$\begin{aligned}\hat{\mathbf{z}}_{s|t} &:= \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t] \\ \hat{\Sigma}_{s|t} &:= \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)\end{aligned}$$

E-step:

$$q^k(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)}) = p_{\theta^k}(\mathbf{z}_{0:T} | \mathbf{x}_{0:T}^{(n)})$$

M-step:

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})} [\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T})]$$

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^{\top}] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_t^{\top}] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^{\top}] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z} | \mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^{\top}]\end{aligned}$$

- These expectations can all be computed via smoothing using θ^k and $\mathbf{x}_{0:T}^{(n)}$
 - To see this, dropping indices k, n for clarity, we have

$$\mathbb{E}[\mathbf{z}_t | \mathbf{x}_{0:T}] = \mathbb{E}[\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_T] = \hat{\mathbf{z}}_{t|T}$$

$$\begin{aligned}\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^{\top} | \mathbf{x}_{0:T}] &= \text{Cov}(\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_T) + \mathbb{E}[\mathbf{z}_t | \mathbf{x}_0, \dots, \mathbf{x}_T] \mathbb{E}[\mathbf{z}_t^{\top} | \mathbf{x}_0, \dots, \mathbf{x}_T] \\ &= \hat{\Sigma}_{t|T} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t|T}^{\top}\end{aligned}$$

$$\mathbb{E}[\mathbf{z}_t \mathbf{z}_{t-1}^{\top} | \mathbf{x}_{0:T}] = \hat{\Sigma}_{t|T} \mathbf{L}_{t-1}^{\top} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t-1|T}^{\top}$$

homework

$$\mathbf{L}_{t-1} = \hat{\Sigma}_{t-1|t-1} \mathbf{A}^{\top} \hat{\Sigma}_{t|t-1}^{-1}$$

M-step (iteration: k)

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

$$\begin{aligned}p(Z_0) &= \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0) \\ p(Z_t | \mathbf{z}_{t-1}) &= \mathcal{N}(\mathbf{A} \mathbf{z}_{t-1}, \mathbf{Q}) \\ p(X_t | \mathbf{z}_t) &= \mathcal{N}(\mathbf{C} \mathbf{z}_t, \mathbf{R})\end{aligned}$$

Model Parameters:
• $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{z \sim q^k(z|x^{(n)})} \left[\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T}) \right]$$

- **Observation.** In the joint log-likelihood

$$p_{\theta}(\mathbf{x}_0, \dots, \mathbf{x}_T, \mathbf{z}_0, \dots, \mathbf{z}_T) = p_{\theta}(\mathbf{z}_0) \prod_{t=0}^T p_{\theta}(\mathbf{x}_t | \mathbf{z}_t) \prod_{t=1}^T p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1}),$$

- $\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0$ only appear in $p_{\theta}(\mathbf{z}_0)$
- $\mathbf{A}, \mathbf{Q}$ only appear in $\prod_{t=1}^T p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1})$
- $\mathbf{C}, \mathbf{R}$ only appear in $\prod_{t=0}^T p_{\theta}(\mathbf{x}_t | \mathbf{z}_t)$

- So the objective of the M-step is separable (as in HMMs), this gives ... (next page)

M-step (iteration: k)

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

$$\begin{aligned}p(\mathbf{Z}_0) &= \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0) \\ p(\mathbf{Z}_t|\mathbf{z}_{t-1}) &= \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q}) \\ p(\mathbf{X}_t|\mathbf{z}_t) &= \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})\end{aligned}$$

Model Parameters:
• $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

$$\theta^{k+1} = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} \left[\log p_{\theta}(\mathbf{x}_{0:T}^{(n)}, \mathbf{z}_{0:T}) \right]$$

- We can therefore decompose M-step into 3 optimization problems (as in HMMs):

M-step $(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$:

$$(\boldsymbol{\pi}_0^{k+1}, \boldsymbol{\Sigma}_0^{k+1}) = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\log p_{\theta}(\mathbf{z}_0)]$$

M-step $(\mathbf{C}, \mathbf{R})$:

$$(\mathbf{C}^{k+1}, \mathbf{R}^{k+1}) = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} \left[\sum_{t=0}^T \log p_{\theta}(\mathbf{x}_t^{(n)} | \mathbf{z}_t) \right]$$

M-step $(\mathbf{A}, \mathbf{Q})$:

$$(\mathbf{A}^{k+1}, \mathbf{Q}^{k+1}) = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} \left[\sum_{t=1}^T \log p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1}) \right]$$

- We will address them one by one next

M-step ($\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0$)

$$p(Z_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

- Since $p_\theta(\mathbf{z}_0)$ is Gaussian, we have:

$$\log p_\theta(\mathbf{z}_0) \propto -\log \det \boldsymbol{\Sigma}_0 - (\mathbf{z}_0 - \boldsymbol{\pi}_0)^\top \boldsymbol{\Sigma}_0^{-1} (\mathbf{z}_0 - \boldsymbol{\pi}_0)$$

- And now we get this:

$$(\boldsymbol{\pi}_0^{k+1}, \boldsymbol{\Sigma}_0^{k+1}) = \operatorname{argmax}_\theta \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\log p_\theta(\mathbf{z}_0)]$$

$$(\boldsymbol{\pi}_0^{k+1}, \boldsymbol{\Sigma}_0^{k+1}) = \operatorname{argmin}_\theta N \log \det \boldsymbol{\Sigma}_0 + \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [(\mathbf{z}_0 - \boldsymbol{\pi}_0)^\top \boldsymbol{\Sigma}_0^{-1} (\mathbf{z}_0 - \boldsymbol{\pi}_0)]$$

M-step ($\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0$)

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

$$(\boldsymbol{\pi}_0^{k+1}, \boldsymbol{\Sigma}_0^{k+1}) = \operatorname{argmin}_{\boldsymbol{\theta}} N \log \det \boldsymbol{\Sigma}_0 + \sum_{n=1}^N \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})}[(\mathbf{z}_0 - \boldsymbol{\pi}_0)^\top \boldsymbol{\Sigma}_0^{-1} (\mathbf{z}_0 - \boldsymbol{\pi}_0)]$$

use the definitions of $\mathbb{E}_k^{(n)}[\mathbf{z}_0]$ and $\mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top]$

$$(\boldsymbol{\pi}_0^{k+1}, \boldsymbol{\Sigma}_0^{k+1}) = \operatorname{argmin}_{\boldsymbol{\theta}} N \log \det \boldsymbol{\Sigma}_0 + N \boldsymbol{\pi}_0^\top \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\pi}_0 + \sum_{n=1}^N \operatorname{tr} \left(\mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top] \boldsymbol{\Sigma}_0^{-1} \right) - 2 \sum_{n=1}^N \boldsymbol{\pi}_0^\top \boldsymbol{\Sigma}_0^{-1} \mathbb{E}_k^{(n)}[\mathbf{z}_0]$$

- Setting the derivative with respect to $\boldsymbol{\pi}_0$ to 0 yields $\boldsymbol{\pi}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0]}{N}$

Use $\boldsymbol{\pi}_0^{k+1}$

$$\boldsymbol{\Sigma}_0^{k+1} = \operatorname{argmin}_{\boldsymbol{\theta}} N \cdot \log \det \boldsymbol{\Sigma}_0 + \sum_{n=1}^N \operatorname{tr} \left(\mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top] \boldsymbol{\Sigma}_0^{-1} \right) - N \cdot \operatorname{tr} \left(\boldsymbol{\pi}_0^{k+1} (\boldsymbol{\pi}_0^{k+1})^\top \boldsymbol{\Sigma}_0^{-1} \right)$$

$m \cdot \log \det \boldsymbol{\Sigma} + \operatorname{tr}(\mathbf{S} \boldsymbol{\Sigma}^{-1})$ is minimized at $\boldsymbol{\Sigma} = \mathbf{S}/m$

$$\boldsymbol{\Sigma}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top]}{N} - \boldsymbol{\pi}_0^{k+1} (\boldsymbol{\pi}_0^{k+1})^\top$$

M-step ($\mathbf{C}, \mathbf{R}$)

$$\mathbb{P}(X_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C}\mathbf{z}_t, \mathbf{R})$$

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

$$(\mathbf{C}^{k+1}, \mathbf{R}^{k+1}) = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{z \sim q^k(z|x^{(n)})} \left[\sum_{t=0}^T \log p_{\theta}(\mathbf{x}_t^{(n)} | \mathbf{z}_t) \right]$$

$p_{\theta}(\mathbf{x}_t^{(n)} | \mathbf{z}_t)$ is Gaussian

$$(\mathbf{C}^{k+1}, \mathbf{R}^{k+1}) = \operatorname{argmin}_{\theta} N(T+1) \log \det \mathbf{R} + \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_{z \sim q^k(z|x^{(n)})} \left[\left(\mathbf{x}_t^{(n)} - \mathbf{C}\mathbf{z}_t \right)^\top \mathbf{R}^{-1} \left(\mathbf{x}_t^{(n)} - \mathbf{C}\mathbf{z}_t \right) \right]$$

use the definitions of $\mathbb{E}_k^{(n)}[\mathbf{z}_t]$ and $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top]$

$$\min_{\theta} N(T+1) \log \det \mathbf{R} + \sum_{n=1}^N \sum_{t=0}^T \left\{ \operatorname{tr} \left(\left(\mathbf{C} \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \mathbf{C}^\top + \mathbf{x}_t^{(n)} (\mathbf{x}_t^{(n)})^\top \right) \mathbf{R}^{-1} - 2 \mathbf{x}_t^{(n)} \mathbb{E}_k^{(n)}[\mathbf{z}_t]^\top \mathbf{C}^\top \mathbf{R}^{-1} \right) \right\}$$

- Setting the derivative with respect to $\mathbf{C}$ to 0 yields

$$\mathbf{C}^{k+1} = \left(\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} \mathbb{E}_k^{(n)}[\mathbf{z}_t]^\top \right) \left(\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \right)^{-1}$$

Use $\mathbf{C}^{k+1}$

$$\min_{\theta} N(T+1) \log \det \mathbf{R} + \sum_{n=1}^N \sum_{t=0}^T \left\{ \operatorname{tr} \left(\mathbf{x}_t^{(n)} (\mathbf{x}_t^{(n)})^\top \mathbf{R}^{-1} - \mathbf{x}_t^{(n)} \mathbb{E}_k^{(n)}[\mathbf{z}_t]^\top (\mathbf{C}^{k+1})^\top \mathbf{R}^{-1} \right) \right\}$$

$$\begin{aligned}\frac{\partial \operatorname{tr}(\mathbf{C}^\top \mathbf{X})}{\partial \mathbf{C}} &= \mathbf{X} \\ \frac{\partial \operatorname{tr}(\mathbf{C}^\top \mathbf{X} \mathbf{C} \mathbf{Y})}{\partial \mathbf{C}} &= \mathbf{X} \mathbf{C} \mathbf{Y} + \mathbf{X}^\top \mathbf{C} \mathbf{Y}^\top\end{aligned}$$

$m \cdot \log \det \mathbf{\Sigma} + \operatorname{tr}(\mathbf{S} \mathbf{\Sigma}^{-1})$ is minimized at $\mathbf{\Sigma} = \mathbf{S}/m$

$$\mathbf{R}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} (\mathbf{x}_t^{(n)})^\top - \mathbf{C}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t] (\mathbf{x}_t^{(n)})^\top}{N(T+1)}$$

M-step ($\mathbf{A}, \mathbf{Q}$)

$$p(Z_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q})$$

$$\begin{aligned}\mathbb{E}_k^{(n)}[\mathbf{z}_t] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_t^\top] \\ \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] &:= \mathbb{E}_{z \sim q^k(z|x^{(n)})}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]\end{aligned}$$

$$(\mathbf{A}^{k+1}, \mathbf{Q}^{k+1}) = \operatorname{argmax}_{\theta} \sum_{n=1}^N \mathbb{E}_{z \sim q^k(z|x^{(n)})} [\sum_{t=1}^T \log p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1})]$$

$\downarrow$ $p_{\theta}(\mathbf{z}_t | \mathbf{z}_{t-1})$ is Gaussian

$$(\mathbf{A}^{k+1}, \mathbf{Q}^{k+1}) = \operatorname{argmin}_{\theta} NT \log \det \mathbf{Q} + \sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_{z \sim q^k(z|x^{(n)})} [(\mathbf{z}_t - \mathbf{A}\mathbf{z}_{t-1})^\top \mathbf{Q}^{-1} (\mathbf{z}_t - \mathbf{A}\mathbf{z}_{t-1})]$$

$\downarrow$ use the definitions of $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top]$ and $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top]$

$$\min_{\theta} NT \log \det \mathbf{Q} + \sum_{n=1}^N \sum_{t=1}^T \left\{ \operatorname{tr} \left(\left(\mathbf{A} \left(\mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top] \right) \mathbf{A}^\top + \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \right) \mathbf{Q}^{-1} - 2 \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] \mathbf{A}^\top \mathbf{Q}^{-1} \right) \right\}$$

- Setting the derivative with respect to $\mathbf{A}$ to 0 yields

$$\mathbf{A}^{k+1} = \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] \right) \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top] \right)^{-1}$$

Use $\mathbf{A}^{k+1}$

$$\min_{\theta} NT \log \det \mathbf{Q} + \sum_{n=1}^N \sum_{t=1}^T \left\{ \operatorname{tr} \left(\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \mathbf{Q}^{-1} - \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] (\mathbf{A}^{k+1})^\top \mathbf{Q}^{-1} \right) \right\}$$

$m \cdot \log \det \mathbf{\Sigma} + \operatorname{tr}(\mathbf{S}\mathbf{\Sigma}^{-1})$ is minimized at $\mathbf{\Sigma} = \mathbf{S}/m$

$$\begin{aligned}\frac{\partial \operatorname{tr}(\mathbf{A}^\top \mathbf{X})}{\partial \mathbf{A}} &= \mathbf{X} \\ \frac{\partial \operatorname{tr}(\mathbf{A}^\top \mathbf{X} \mathbf{A} \mathbf{Y})}{\partial \mathbf{A}} &= \mathbf{X} \mathbf{A} \mathbf{Y} + \mathbf{X}^\top \mathbf{A} \mathbf{Y}^\top\end{aligned}$$

$$\mathbf{Q}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] - \mathbf{A}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_t^\top]}{NT}$$

Summary: EM for LDSs (iteration: k)

E-step

Given θ^k , for each $\mathbf{x}_{0:T}^{(n)}$, use **Kalman Filter** & **Smoothing** to compute:

- $\mathbb{E}_k^{(n)}[\mathbf{z}_t] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t | \mathbf{x}_{0:T}^{(n)}]$
- $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t \mathbf{z}_t^\top | \mathbf{x}_{0:T}^{(n)}]$
- $\mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] := \mathbb{E}_{\mathbf{z} \sim q^k(\mathbf{z}|\mathbf{x}^{(n)})} [\mathbf{z}_t \mathbf{z}_{t-1}^\top | \mathbf{x}_{0:T}^{(n)}]$

← add indices n, k

M-step

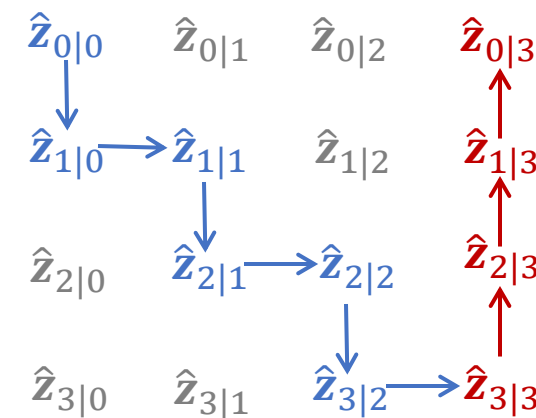
Update parameters:

- $\boldsymbol{\pi}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0]}{N}$
- $\boldsymbol{\Sigma}_0^{k+1} = \frac{\sum_{n=1}^N \mathbb{E}_k^{(n)}[\mathbf{z}_0 \mathbf{z}_0^\top]}{N} - \boldsymbol{\pi}_0^{k+1} (\boldsymbol{\pi}_0^{k+1})^\top$
- $\mathbf{C}^{k+1} = \left(\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} \mathbb{E}_k^{(n)}[\mathbf{z}_t]^\top \right) \left(\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] \right)^{-1}$
- $\mathbf{R}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbf{x}_t^{(n)} (\mathbf{x}_t^{(n)})^\top - \mathbf{C}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t] (\mathbf{x}_t^{(n)})^\top}{N(T+1)}$
- $\mathbf{A}^{k+1} = \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_{t-1}^\top] \right) \left(\sum_{n=1}^N \sum_{t=1}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top] \right)^{-1}$
- $\mathbf{Q}^{k+1} = \frac{\sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_t \mathbf{z}_t^\top] - \mathbf{A}^{k+1} \sum_{n=1}^N \sum_{t=0}^T \mathbb{E}_k^{(n)}[\mathbf{z}_{t-1} \mathbf{z}_t^\top]}{NT}$

- $\hat{\mathbf{z}}_{s|t} := \mathbb{E}[\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t]$
- $\hat{\boldsymbol{\Sigma}}_{s|t} := \text{Cov}(\mathbf{z}_s | \mathbf{x}_0, \dots, \mathbf{x}_t)$
- $\mathbf{L}_{t-1} := \hat{\boldsymbol{\Sigma}}_{t-1|t-1} \mathbf{A}^\top \hat{\boldsymbol{\Sigma}}_{t|t-1}^{-1}$

Kalman Filter & **Smoothing** can compute

- $\mathbb{E}[\mathbf{z}_t | \mathbf{x}_{0:T}] = \hat{\mathbf{z}}_{t|T}$
 - $\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^\top | \mathbf{x}_{0:T}] = \hat{\boldsymbol{\Sigma}}_{t|T} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t|T}^\top$
 - $\mathbb{E}[\mathbf{z}_t \mathbf{z}_{t-1}^\top | \mathbf{x}_{0:T}] = \mathbf{L}_{t-1} \hat{\boldsymbol{\Sigma}}_{t|T} + \hat{\mathbf{z}}_{t|T} \hat{\mathbf{z}}_{t-1|T}^\top$
- via **forward** & **backward** passes



Model Parameters:

- $\theta := (\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0, \mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{R})$

$$p(\mathbf{Z}_0) = \mathcal{N}(\boldsymbol{\pi}_0, \boldsymbol{\Sigma}_0)$$

$$p(\mathbf{Z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{A} \mathbf{z}_{t-1}, \mathbf{Q})$$

$$p(\mathbf{X}_t | \mathbf{z}_t) = \mathcal{N}(\mathbf{C} \mathbf{z}_t, \mathbf{R})$$